Module 1 National Academic Standards (including Oklahoma State Standards):

National Academic Standards Next Generation Science Standards Common Core Standards for ELA and Math		
Science	**Science: SEP/DCI/CCC**	**ELA**
4.Energy **4-PS3-1** **4-PS3-2** **4.Earth's Systems: Processes that Shape the Earth** **4-ESS2-2** **4-ESS3-2** **5.Earth's Systems** **5-ESS2-2** **MS.Structure and Properties of Matter** **MS-PS1-1** **MS-PS1-4** **HS.Earth's Systems** **HS-ESS2-5**	**Science & Engineering Practice** ● Asking questions and defining problems ● Planning and carrying out investigations ● Analyzing and interpreting data ● Constructing explanations and designing solutions ● Developing and using models ● Obtaining, evaluating and communicating information ● Engaging in argument from evidence **Disciplinary Core Ideas** ● PS3.A: Definitions of energy ● PS3.B: Conservation of energy and energy transfer ● ESS2.A: Earth materials and systems ● ESS2.C: The roles of water in Earth's surface processes ● ESS2.D: Weather and climate ● ESS3.A: Natural resources ● ESS3.C: Human impacts on Earth systems ● PS1.A: Structure and properties of matter ● PS3.A: Definitions of energy **Crosscutting Concepts** ● Patterns ● Cause and effect ● Scale, proportion, and quantity ● Scientific knowledge assumes an order and consistency in natural systems ● Systems and system models ● Energy and matter ● Stability and change	**Writing Standards K–5 (Grades 4 & 5)** ***Research to Build and Present Knowledge*** **Sub-standards 7-9** **Speaking and Listening Standards K–5 (Grades 4 & 5)** ***Comprehension and Collaboration*** **Sub-standard 1.a-d** ***Presentation of Knowledge and Ideas*** **Sub-standard 4-6** **Writing Standards 6–12** ***Research to Build and Present Knowledge*** **Sub-standard 7** **Speaking and Listening Standards 6–12**

		Comprehension and Collaboration **Sub-standard 1.a-d** ***Presentation of Knowledge and Ideas*** **Sub-standards 4-6**
Oklahoma Academic Standards		
Science	**Science: SEP/DCI/CCC**	**ELA**

Energy (PS3) **4.PS3.1** **4.PS3.2** **Earth's Place in the Universe (ESS1)** **5.ESS2.1** **5.ESS2.2** **Matter and Its Interactions (PS1)** **6.PS1.4** **Earth's Systems (ESS2)** **6.ESS2.1** **6.ESS2.4** **HS Earth and Space Science (ES)** **Earth Systems (ESS2)** **ES.ESS2.5** **HS Environmental Science (EN)** **EN.ESS2.5**	**Science & Engineering Practice** ● Constructing explanations ● Planning and carrying out/conducting investigations ● Developing and using models **Disciplinary Core Ideas** ● The faster a given object is moving, the more energy it possesses ● Energy can be moved from place to place by moving objects or through sound, light, or electric currents. ● Energy is present whenever there are moving objects, sound, light, or heat. ● When objects collide, energy can be transferred from one object to another, thereby changing their motion. In such collisions, some energy is typically also transferred to the surrounding air; as a result, the air gets heated and sound is produced. ● Earth's major systems are the geosphere, hydrosphere, atmosphere, and biosphere. ● Nearly all of Earth's available water is in the ocean. ● Most freshwater is in glaciers or underground; only a tiny fraction is in streams, lakes, wetlands, and the atmosphere. ● Gases and liquids are made of molecules or inert atoms that are moving about relative to each other. In a liquid, the molecules are constantly in contact with others; in a gas, they are widely spaced except when they happen to collide. In a solid, atoms are closely spaced and may vibrate in position but do not change relative locations. ● The changes of state that occur with variations in temperature or pressure can be described and predicted using these models of matter. ● The changes of state that occur with variations in temperature or pressure can be described and predicted using these models of matter. ● All Earth processes are the result of energy flowing and matter cycling within and among the planet's systems. This energy is derived from the sun and Earth's hot interior. The energy that flows	**Standard 1: Listening and Speaking** ***All listed sub-standards*** **(Grades 4-12)** **Standard 3: Critical Reading and Writing** ***Reading sub-standards 6 through the last sub-standard*** **(Grades 4-8)** ***Reading sub-standards 5 through the last sub-standard*** **(Grades 9-12)** ***Writing sub-standard 2 through the last sub-standard*** **(Grades 4-12)** **Standard 6: Research** ***All listed sub-standards*** **(Grades 4-12)** **Standard 7: Multimodal Literacies** ***All listed sub-standards*** **(Grades 4-12)**

	and matter that cycles produces chemical and physical changes in Earth's materials ● Water continually cycles among land, ocean, and atmosphere via transpiration, evaporation, condensation, and crystallization, and precipitation, as well as downhill flows on land. ● Global movements of water and its changes in form are propelled by sunlight and gravity. ● The abundance of liquid water on Earth's surface and its unique combination of physical and chemical properties are central to the planet's dynamics. These properties include water's exceptional capacity to absorb, store, and release large amounts of energy; transmit sunlight; expand upon freezing; dissolve and transport materials; and lower the viscosities and melting points of rocks. **Crosscutting Concepts** ● Patterns ● Energy and matter ● System and system models ● Cause and effect ● Stability and change ● Structure and function	